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# A robotic grinding parameter optimization approach considering tool wear to guarantee material removal accuracy and surface quality

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ABSTRACT

High-precision curved parts grinding must meet strict material removal control and surface quality constraint at the same time. In actual grinding process, the tool wear will cause its material removal ability to gradually decay and simultaneously induce changes in workpiece surface roughness, while existing studies have not considered the dynamic impact of tool wear to simultaneously guarantee material removal accuracy and surface quality. This paper proposes a robotic grinding parameter optimization method considering tool wear to guarantee material removal accuracy while maintain surface quality. Firstly, after quantifying the tool wear over usage time, the material removal depth coefficient under different process parameters at different times is predicted by multivariate Gaussian process regression, and a material removal model considering tool wear is developed. Secondly, a surface roughness prediction model is constructed in combination with the tool wear degree to characterize the influence of tool wear on surface quality. On this basis, a robotic grinding process parameter optimization algorithm considering tool wear is developed, which achieves accurate compensation of material removal accuracy by adjusting process parameters, while ensuring that the surface quality meets the requirements. Finally, experimental verification results show that compared with the robotic grinding process parameter optimization method without considering tool wear and surface roughness constraint, the proposed method significantly improves the material removal accuracy and surface quality of the curved parts after grinding.

## 1. Introduction

Complex curved surface components serve as critical parts in aero-engines and typically feature intricate geometries, tight machining tolerances, and stringent requirements for contour accuracy and surface roughness. To meet these demanding specifications, compliant grinding processes are often employed [1–4]. While manual grinding remains the dominant grinding method for such complex surfaces, it suffers from limitations in both efficiency and precision. In contrast, robotic grinding offers distinct advantages, including cost-effectiveness, operational flexibility, and a large working space, making it increasingly prevalent in industrial applications [5–8].

In robotic grinding, the material removal amount can be adjusted by accurately controlling contact force. Currently, there are two main forms of robotic active force control in grinding process: robot manipulator motion-based force control [9–11] and active force-controlled end-effector [12,13]. Because of the low inertia and high response speed, the force-controlled end-effector method has higher control precision than the method of robot manipulator motion-based force control, and it is especially appropriate for high-precision robot grinding [14,15]. For example, Mohammad et al. [16] used voice coil motor to dynamically adjust contact force by controlling the tool axial displacement. Li et al. [17] and Chen et al. [18] developed a servo-driven force-controlled end-effector that improved accuracy and surface quality in complex curved part grinding. Capitalizing on the high-precision characteristics of force-controlled end-effector, this paper uses it to perform grinding operations on curved parts.

To achieve high-precision material removal of curved components, establishing an accurate material removal depth (MRD) model is essential [19–21]. Current researches [17,22] primarily focuses on developing composite MRD models that incorporate multiple process parameters, including: robot feedrate, spindle speed, contact force, and

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local curvature at the contact point. To ensure the uniform MRD, Li et al. [17] planned the force at different curvature locations according to the MRD model at a constant feedrate, and verified the effectiveness through grinding experiments. Ge et al. [23] adaptively adjusted the robot feedrate and the grinding tool rotation speed according to the required MRD at each position point to improve welding seam grinding accuracy. Kakinuma et al. [24] developed a way to achieve uniform MRD by simultaneously controlling the spindle and force according to the tool feedrate. However, in the above methods, when grinding process parameters are planned according to the MRD model, the grinding tool wear is not considered, and the MRD coefficient is treated as a fixed value. In fact, as the usage time of the grinding tool increases, the abrasive particles on the tool surface are prone to wear, causing the material removal capacity to continue to decrease [25]. The actual material removal of the part will be smaller than the theoretical value, resulting in under-grinding and affecting the final contour accuracy.

In addition to ensuring the material removal accuracy, the surface roughness after grinding must also meet the requirements. Existing studies [26–28] have mainly concentrated on establishing comprehensive surface roughness models that integrate key process parameters as robot feedrate, spindle speed, and force. In order to obtain better grinding surface quality, Nguyen et al. [29] investigated the effects of process parameters like feedrate and inclination angle on surface roughness and used multi-response surface optimization to establish optimal processing conditions. Zhao et al. [30] employed response surface methodology to enhance blade surface finish by minimizing roughness through the optimization of grinding parameters, including particle size, contact force, and robot feedrate. Yang et al. [31] used orthogonal experiments to evaluate the effects of rotation speed and robot feedrate on surface roughness, and used improved genetic algorithm to optimize these parameters to ensure surface quality. However, in the above methods, the grinding tool wear is not also considered. In the actual grinding process, tool wear will affect the roughness of workpiece [25]. Therefore, the process parameter optimization results derived from the surface roughness model constraint that does not consider tool wear will deteriorate with the usage time.

Therefore, studying the tool wear during grinding process is of considerable importance. In order to monitor tool wear, an acoustic emission-driven predictive model was established by Xu et al. [32] to evaluate tool wear state in titanium alloy grinding. Cheng et al. [33] adopted multi-sensor fusion method integrating acoustic and current signals and then applied Bayesian network model to predict the tool wear status. Caesarendra et al. [34] proposed a monitoring method based on convolutional neural network to assess belt wear utilizing vibration and force signatures, elucidating the influence of vibration directionality and force characteristics on tool wear state during grinding process. However, the above studies mainly focus on monitoring the wear state itself and do not analyze in depth the influence of grinding process parameters on the evolution of tool wear. To address this issue, Cao et al. [35] and Li et al. [36] conducted tool wear experiments under different feedrate and normal force process parameters, revealing the influence of feedrate and normal force on tool wear.

Furthermore, to quantitatively study the relationship between tool wear and material removal, Zhang et al. [37] proposed a method to predict the time-varying Preston coefficient using Gaussian process regression. However, this method assumes that material removal is determined only by instantaneous force and processing time, so it is only applicable to constant force grinding scenarios but is difficult to apply to variable force grinding processes. Under variable force conditions, historical contact force has a significant impact on instantaneous wear behavior, and the material removal process shows obvious force history correlation and is affected by the cumulative effect of previous force action. Therefore, it is urgent to establish wear modeling and compensation strategies that take into account the effects of force history.

Taking into account various influencing factors, it is crucial to control material removal accuracy and surface roughness in robotic grinding tasks. Current process optimization research mainly considers only single performance indicators such as MRD or surface roughness, and there is also a lack of research that simultaneously examines the effects of force, feedrate, tool wear, and spindle speed on MRD and surface roughness. Therefore, this paper developes a process optimization approach that simultaneously considers material removal accuracy and surface quality constraints. The main contributions of this work are outlined as follows: (1) MRD coefficient are predicted through multivariate Gaussian process regression under different process parameters at different times, and a material removal model considering tool wear is established. (2) A surface roughness model that considers tool wear is developed to characterize the influence of tool wear on surface roughness. (3) A robot grinding parameter optimization method that simultaneously considers material removal accuracy and surface quality is proposed. By adjusting the process parameters, the material removal depth error caused by tool wear can be accurately compensated while ensuring that the surface quality meets the requirement. Experimental results indicate that the developed parameter optimization approach improves the material removal accuracy and surface quality of the workpiece.

The remainder of this paper is structured as follows. Section 2 details the tool wear quantification and prediction of MRD coefficient using multivariate Gaussian process regression approach. A surface roughness prediction model is constructed to characterize the influence of tool wear on surface quality in Section 3. A robotic grinding parameter optimization approach considering tool wear to ensure MRD and surface quality is proposed in Section 4. Sections 5 and 6 present the experiments and conclusions, respectively.

## 2. Material removal modeling considering tool wear

### 2.1. Theoretical material removal model without considering tool wear

To attain the target MRD, establishing a precise material removal model is crucial. This paper adopts a theoretical material removal model of compliant grinding based on the Preston equation [17] as follows

$$h = \frac{k_p n}{v}\sqrt{F\left(\frac{1}{r_t}+\frac{1}{r_w}\right)} \tag{1}$$

where $h$ represents the MRD. $k_p$ represents the MRD coefficient, which is treated as constant when tool wear is not considered. $n$ represents the spindle speed, $v$ represents the robot feedrate, $F$ represents the force, $r_t$ represents the tool radius, and $r_w$ represents the curvature radius of the workpiece. Fig. 1 illustrates the tool-workpiece contact.

In existing studies [17,22], the MRD coefficient $k_p$ is modeled as a constant value and independent of tool wear and grinding parameters. However, during extended grinding process, abrasive particle wear and detachment from grinding tool surface progressively reduce material removal capacity. This leads to actual removal amount falling below theoretical values, causing under-grinding and compromising material

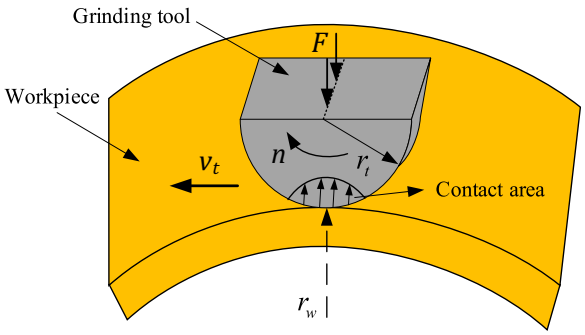



**Fig. 1.** Schematic of the tool-workpiece contact.

removal accuracy.

Fig. 2 presents the measured micromorphology of abrasive grains on tool surface at varying usage times using the laser scanning confocal microscope VK-X200K. The tool consists of a rubber base with diamond abrasive. As presented in Fig. 2, with prolonged usage, the height of abrasive grains on the grinding tool surface progressively diminishes. This reduction in abrasive grain height reduces the cutting ability of the grinding tool, ultimately leading to a decrease in material removal capacity [25]. In the initial wear phase, tool wears rapidly, resulting in a fast reduction of material removal rate. After entering the stable wear phase, the wear rate slows, and the material removal rate slowly decreases. In the accelerated wear phase, the abrasive grains are essentially completely worn flat, and the material removal capacity stabilizes but remains low. At this stage, the material removal rate significantly decreases, and the tool loses its practical value.

Therefore, to accurately obtain the material removal models across different stages, studying the changing process of tool wear and corresponding effects of grinding parameters on MRD coefficient $k_p$ is necessary.

### 2.2. Tool wear quantification and MRD coefficient prediction

#### 2.2.1. Tool wear quantification

An experiment setup, illustrated in Fig. 3, is developed to study the correlation between tool wear and grinding process parameters. The compliant force control end-effector [5] is mounted at the robot flange to achieve closed-loop control of the force. The desired spindle speed is realized through adjusting the current of the electric spindle, and the feedrate control of tool is achieved through the turntable.

First, several new grinding heads are used for grinding the cylindrical bar, where the base of the tool is rubber and the abrasive is diamond, and the material of the bar is TC4. Then the bar is ground under different combinations of process parameters of contact force and feedrate (Table 1). During robot grinding, the spindle operates at a fixed rotational speed. The material removal mass of the bar is measured after each ground of grinding. To minimize the influence of initial rapid wear stage of the grinding head, the first circle is regarded as the pre-wear of the grinding head, and the data from the first circle are not included in the experimental results.

Fig. 4 presents the tool wear conditions under different contact forces. Fig. 4(a) illustrates the material removal mass $m$ changing with the number of turns $x$ under different contact forces at the same feedrate of 1 mm/s. After obtaining the mass of removed material for each circle around the cylindrical bar $\Delta m$, the MRD $h$ of the cylindrical bar in each grinding turn can be calculated by $h = \psi/\rho$ ($\psi$ is the removed material mass per unit contact area, and $\rho$ is the density of the workpiece material). Then, the MRD coefficient of tool after grinding a turn of cylindrical bar can be calculated according to Eq. (1). When the tool-workpiece feedrate is 1 mm/s, since the bar radius is 5 mm, the time for each turn of the bar to be grinded is $10\pi$ s. Then the relationship between grinding turns $x$ and time $t$ is $x = t/10\pi$. Therefore, the relationship between $\psi$ and $x$ can be converted into the relationship between $k_p$ and $t$.

Fig. 4(b) depicts the variation of $k_p$ over time under different contact forces at the same feedrate of 1 mm/s. It shows that in the early stage of tool grinding, the tool wears faster and $k_p$ decreases faster. This is because the abrasive grains on the tool are prone to falling off and breaking at the early stage of tool usage. After a period of grinding, the tool wear enters a stable period and $k_p$ decreases slowly. When the wear reaches a certain level, $k_p$ hardly changes, and the tool loses its practical value.

Fig. 5 depicts the tool wear conditions under different feedrate, indicating that the change of tool feedrate has almost no effect on the change of MRD coefficient, so the change of tool feedrate has little effect on tool wear. In order to guarantee surface quality of grinding, the spindle speed $n$ is set to a relatively high value of 16,000 rpm, while the feedrate $v$ is set to a relatively small value (from 1 to 4 mm/s) in experiments. Therefore, the tool feedrate is much smaller than the feedrate caused by the spindle speed. Consequently, the influence of tool feedrate on tool wear is negligible within a certain process constraint range during grinding.

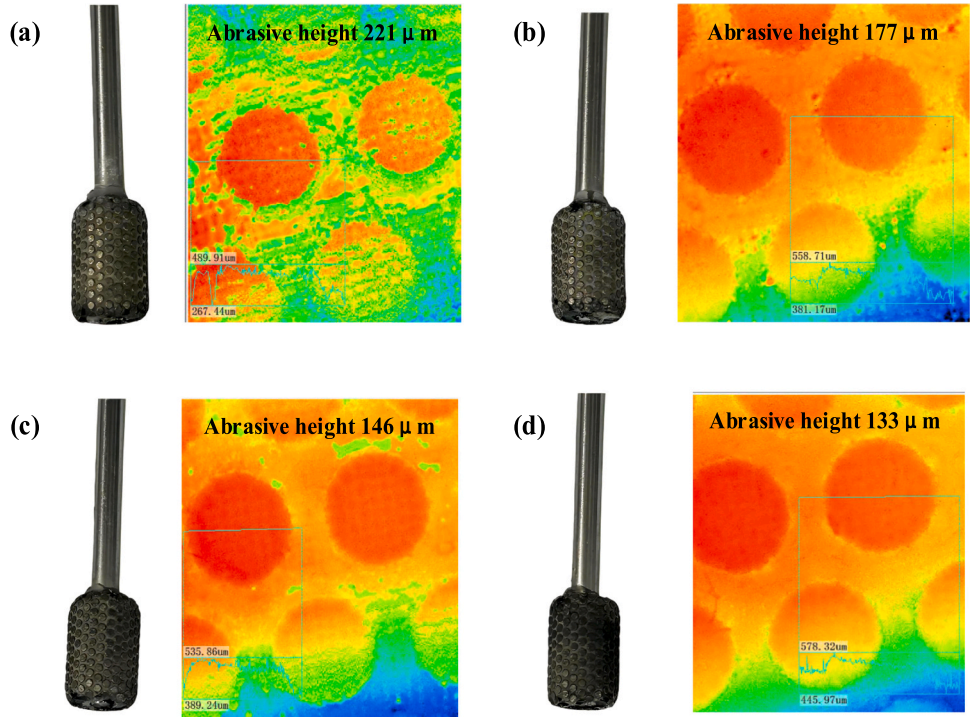



**Fig. 2.** Evolution of abrasive grain morphology with tool usage time ($n$ = 16,000 rpm, $F$ = 3 N, $v$ = 1 mm/s): (a) new tool; (b) usage time at 3 min; (c) usage time at 6 min; (d) usage time at 10 min.

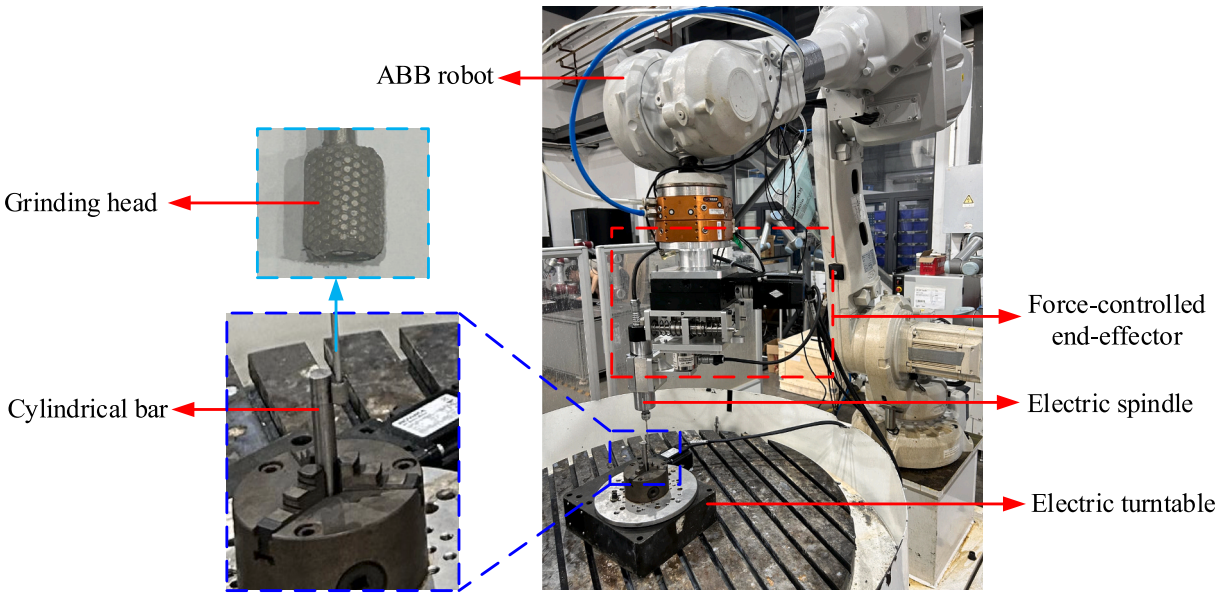



**Fig. 3.** Experimental platform of material removal depth coefficient calibration.

**Table 1**
Experimental parameters of grinding process for studying tool wear.

| Influencing factors | Parameters |
|---|---|
| Contact force (N) | 1, 2, 3, 4, 5 |
| Feedrate (mm/s) | 1, 2, 3, 4 |
| Spindle speed (rpm) | 16,000 |
| Belt mesh | 600 |
| Bar radius (mm) | 5 |

*2.2.2. Prediction of MRD coefficient with the consideration of tool wear*

Since the MRD coefficient varies with time and process parameters, accurately predicting its value under different usage time and grinding process parameters is essential for achieving the desired MRD.

According to the experimental results of subsection 2.2.1, both grinding contact force and grinding time influence the MRD coefficient, and the feedrate has almost no effect on the MRD coefficient. Therefore, this paper only studies the change of MRD coefficient $k_p$ with grinding contact force $F$ and grinding time $t$.

To describe the tool wear conditions accurately, tool wear degree $w(t)$ is defined as the percentage of reduction in MRD coefficient, as expressed by the following formula

$$w(t) = 1 - \frac{k_p(t)}{k_{p0}} \tag{2}$$

where $k_p(t)$ and $k_{p0}$ are the real-time material removal depth coefficient and the initial tool's material removal depth coefficient, respectively.

During the entire grinding, the tool wear process can be modeled as

$$w(t) = w(w_0, F(1), F(2), \ldots, F(t)) \tag{3}$$

where $w_0$ is the initial tool wear degree, which is defined as zero for a new tool, and $F(t)$ is the contact force at time $t$. Tool wear rate $r(t)$ is defined as the percentage reduction in MRD coefficient per unit grinding time, which is related to tool wear degree and contact force at the current moment $t$. Therefore, Eq. (3) is expressed as

$$w(t) = w(t-1) + r(t) = w(t-1) + r(w(t-1), F(t)) \tag{4}$$

Based on Eq. (4), the tool wear rate is integrated to obtain the tool wear degree. After obtaining the tool wear degree $w(t)$, the real-time material removal depth coefficient $k_p(t)$ is calculated using Eq. (2).

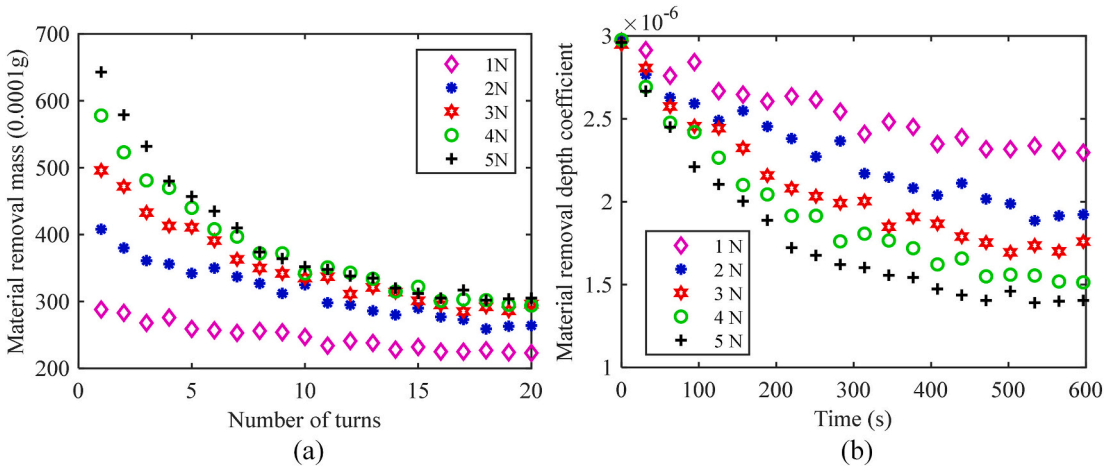



**Fig. 4.** Tool wear diagram: (a) material removal mass changes with number of turns (at the same feedrate of 1 mm/s and different contact forces); (b) material removal coefficient changes with time (at the same feedrate of 1 mm/s and different contact forces).

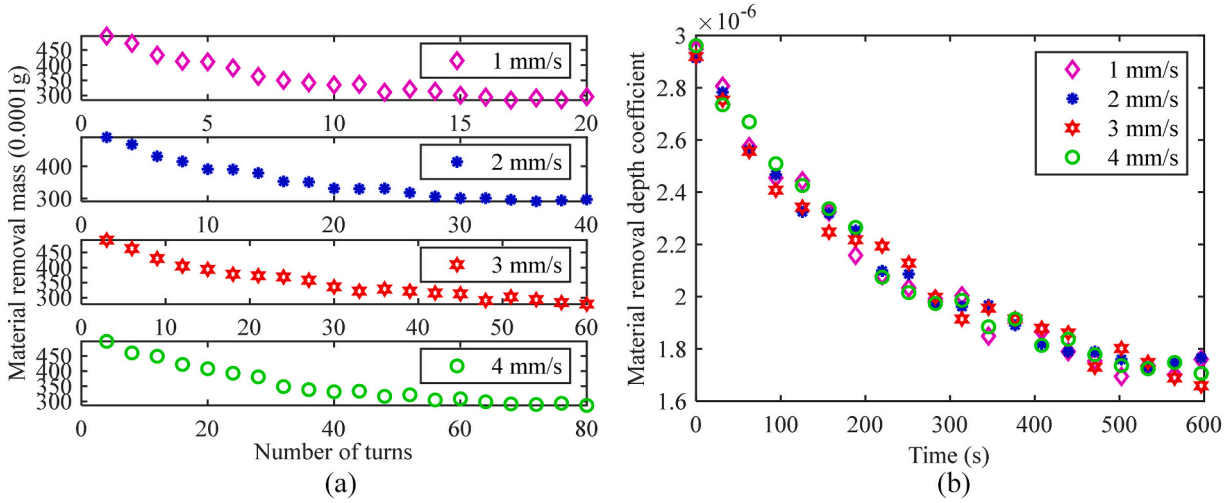



**Fig. 5.** Tool wear diagram: (a) material removal mass changes with number of turns (at the same contact force of 3 N and different feedrates); (b) material removal coefficient changes with time (at the same contact force of 3 N and different feedrates).

Therefore, the material removal model considering time-varying tool wear is as follows

$$h = \frac{k_p(t)n}{v}\sqrt{F\left(\frac{1}{r_t}+\frac{1}{r_w}\right)} \tag{5}$$

Gaussian Process Regression (GPR) is a kernel-based, non-parametric Bayesian modeling method that employs Gaussian process priors for regression tasks. Its flexible kernel functions and probabilistic prediction outputs make it particularly effective for time series analysis applications [38,39]. Therefore, this paper constructs multivariate Gaussian process regression model to predict the real-time tool wear rate $r(t)$ and then calculates the MRD coefficient $k_p(t)$ with the consideration of the tool wear.

The training set $E = \{([w_1, F_1]^{\mathrm{T}}, r_1), ([w_2, F_2]^{\mathrm{T}}, r_2), \ldots, ([w_n, F_n]^{\mathrm{T}}, r_n)\}$ is composed of 95 sets of data obtained from the grinding experiments in Section 2.2.1, where $w_n$, $F_n$, and $r_n$ correspond to the $n$-th tool wear degree, contact force, and tool wear rate in the data set, respectively. The GPR process is generated from

$$\boldsymbol{y} = f(\boldsymbol{x}) + \boldsymbol{\varepsilon} \tag{6}$$

where $\boldsymbol{x}$ is the matrix of different tool wear degree $w$ and contact forces $F$, $\boldsymbol{y}$ is the corresponding measured tool wear rate vector, $\boldsymbol{\varepsilon}$ is the noise that conforms to a Gaussian distribution $N(0,\sigma_n{}^2)$, and $\sigma_n{}^2$ is covariance of the noise. $f(\boldsymbol{x})$ is the function to be predicted, which satisfies the following Gaussian distribution

$$f(\boldsymbol{x}) \sim N(m(\boldsymbol{x}), k(\boldsymbol{x},\boldsymbol{x}^{'})) \tag{7}$$

where $m(\boldsymbol{x}) = E[f(\boldsymbol{x})]$ is mean function, $k(\boldsymbol{x},\boldsymbol{x}^{'}) = E[(f(\boldsymbol{x}) - m(\boldsymbol{x}))(f(\boldsymbol{x}^{'}) - m(\boldsymbol{x}^{'}))]$ is covariance function, and $\boldsymbol{x},\boldsymbol{x}^{'}$ are input vectors of the training set.

The joint probability density distribution of training data and test data is

$$\begin{bmatrix}\boldsymbol{y}\\ f^{*}\end{bmatrix} \sim N\left(\begin{bmatrix}m(\boldsymbol{x})\\ m(\boldsymbol{x}^{*})\end{bmatrix}, \begin{bmatrix}k(\boldsymbol{x},\boldsymbol{x}) + \sigma_n{}^2\boldsymbol{I} & k(\boldsymbol{x},\boldsymbol{x}^{*})\\ k(\boldsymbol{x}^{*},\boldsymbol{x}) & k(\boldsymbol{x}^{*},\boldsymbol{x}^{*})\end{bmatrix}\right) \tag{8}$$

where $\boldsymbol{x}^{*}$ is new input vector of the test set. Finally, the predicted value $m^{*}$ of the tool wear rate $r(t)$ corresponding to the new input data are

$$m^{*} = m(\boldsymbol{x}^{*}) + k\left(\boldsymbol{x}^{*},\boldsymbol{x}\right)\left(k(\boldsymbol{x},\boldsymbol{x}) + \sigma_n{}^2\boldsymbol{I}\right)^{-1}(\boldsymbol{y} - m(\boldsymbol{x})) \tag{9}$$

In GPR, the nonlinear mapping between grinding force and time is established through the kernel function, and the material removal coefficient is accurately predicted. The covariance function is designed as a composite kernel function of the exponential kernel and the periodic kernel as follows

$$k\left(\boldsymbol{x},\boldsymbol{x}^{'}\right) = \theta_1{}^2\exp\left(-\frac{(\boldsymbol{x}-\boldsymbol{x}^{'})^2}{2\theta_2{}^2}\right) + \theta_3{}^2\exp\left(-\frac{2}{\theta_4{}^2}\sin^2\left(\frac{\pi(\boldsymbol{x}-\boldsymbol{x}^{'})}{T}\right)\right) \tag{10}$$

where $\theta_1, \theta_2, \theta_3, \theta_4$ and $T$ are hyperparameters, and the optimal hyperparameters are determined by maximizing the marginal log-likelihood function based on the given training set $E$ using the Sequential Quadratic Programming (SQP) method. Specifically, the optimization is performed by maximizing the following formula

$$\ln p(\boldsymbol{y}|\boldsymbol{x},\theta) = -\frac{1}{2}\ln|k + \sigma_n{}^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{y}^{\mathrm{T}}(k + \sigma_n{}^2\boldsymbol{I})^{-1}\boldsymbol{y} - \frac{n}{2}\ln(2) \tag{11}$$

where $n$ represents the size of training set $E$.

To verify the superiority of the proposed MRD coefficient prediction method considering the influence of historical contact force, the proposed method is compared with the GPR model proposed in reference [37], and grinding experiments are carried out on the cylindrical bar (bar radius is 5 mm) again. The actual MRD coefficients under different contact forces and different times are calculated (the calculation method is described in Reference [17]), and then compared with the MRD coefficients predicted by the proposed method and the comparison method.

According to the experimental results shown in Fig. 6, when the contact force is 1.5 N, the root mean square errors (RMSEs) of the MRD coefficient ($k_p(t)$ in Eq. (5)) prediction using the proposed method and the comparison method [37] are 0.038 × $10^{-6}$ and 0.042 × $10^{-6}$, respectively; the mean absolute errors (MAEs) are 0.034 × $10^{-6}$ and 0.036 × $10^{-6}$, respectively; and the coefficient of determination $R^2$ are 0.941 and 0.936, respectively. When the contact force is 3.5 N, the RMSEs of the MRD coefficient prediction using the proposed method and the comparison method are 0.043 × $10^{-6}$ and 0.052 × $10^{-6}$, respectively; the MAEs are 0.033 × $10^{-6}$ and 0.039 × $10^{-6}$, respectively; and the $R^2$ are 0.953 and 0.946, respectively. Therefore, for constant contact forces of 1.5 N and 3.5 N at different times, both the proposed method and the comparison method can effectively predict the MRD coefficient, and the prediction results are almost identical. Under the variable contact force condition (Table 2), the RMSEs of the MRD coefficient prediction using the proposed method and the comparison method are 0.062 × $10^{-6}$ and 0.15 × $10^{-6}$, respectively; the MAEs are 0.056 × $10^{-6}$ and 0.11 × $10^{-6}$, respectively; and the $R^2$ are 0.943 and 0.777,

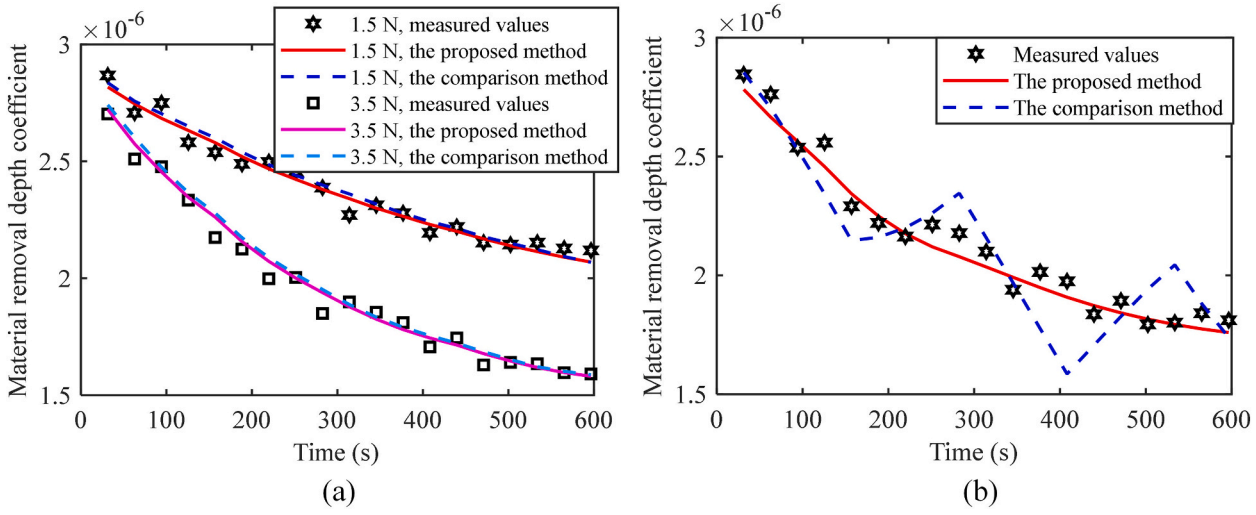



**Fig. 6.** Comparison of MRD coefficients predicted using the proposed method and the comparison method [37] with actual measured values: (a) contact force of 1.5 N and 3.5 N at different times; (b) variable contact force at different times.

respectively. The prediction accuracy of the proposed method is significantly higher than that of the comparison method. The MRD coefficients predicted by the proposed method and actual measured values corresponding to different grinding stage are shown in Table 2. Therefore, the effectiveness and robustness of the proposed material removal depth coefficient prediction method considering the influence of historical contact force is verified.

The key innovation of the proposed method for predicting MRD coefficient lies in explicitly modeling the dynamic influence of historical contact force and tool wear degree on the current tool wear rate. Specifically, the proposed GPR model takes the tool wear degree at the previous time step $w(t\text{-}1)$ and the current contact force $F(t)$ as inputs to predict the current tool wear rate $r(t)$. The tool wear rate is then integrated over time to obtain the tool wear degree, ultimately yielding the time-varying MRD coefficient. In contrast, existing method based on GPR [37] only predict the MRD coefficient directly from instantaneous force and time, implicitly assuming that material removal is solely affected by the current force. This model has applicability in constant force grinding, but it has limitations in variable force grinding processes because it ignores the dynamic influence of historical contact force on instantaneous tool wear behavior.

The proposed method uses the Eq. (4) to effectively capture the cumulative effect of force history on tool wear, thus making it applicable to the prediction of material removal under time-varying force conditions. This method addresses the shortcomings of existing research in modeling variable force grinding and provides an effective means for accurate prediction of material removal behavior in actual robotic grinding processes.

**Table 2**
Comparison of MRD coefficients predicted by the proposed method and actual measured values under variable contact force.

| Number of turns | Force (N) | Predicted material removal depth coefficient ($\times10^{-6}$) | Measured material removal depth coefficient ($\times10^{-6}$) |
|---|---|---|---|
| 1 | 1 | 2.7822 | 2.8455 |
| 2 | 2 | 2.6658 | 2.7619 |
| 3 | 3 | 2.5634 | 2.5373 |
| 4 | 4 | 2.4578 | 2.5590 |
| 5 | 5 | 2.3434 | 2.2906 |
| 6 | 4 | 2.2495 | 2.2207 |
| 7 | 3 | 2.1756 | 2.1634 |
| 8 | 2 | 2.1202 | 2.2138 |
| 9 | 1 | 2.0788 | 2.1776 |
| 10 | 2 | 2.0355 | 2.1000 |
| 11 | 3 | 1.9917 | 1.9392 |
| 12 | 4 | 1.9488 | 2.0145 |
| 13 | 5 | 1.9081 | 1.9751 |
| 14 | 4 | 1.8730 | 1.8368 |
| 15 | 3 | 1.8423 | 1.8934 |
| 16 | 2 | 1.8158 | 1.7938 |
| 17 | 1 | 1.7933 | 1.8012 |
| 18 | 2 | 1.7748 | 1.8409 |
| 19 | 3 | 1.7598 | 1.8115 |

## 3. Surface roughness modeling considering tool wear

Tool wear not only changes MRD, but also affects the workpiece surface roughness [25]. Quantitative research on the correlation mechanism between tool wear and surface roughness offers a theoretical foundation for subsequent process parameter optimization.

### 3.1. Empirical formula of surface roughness for robotic grinding

Surface roughness serves as the primary metric for assessing surface quality. To enable precise roughness control, a predictive model must be developed that correlates surface roughness with key grinding parameters.

This paper first studies the correlation between surface roughness and key process parameters after grinding with a new grinding head through single factor experiments (Fig. 7). Experimental results indicate that, within a certain range, surface roughness decreases as spindle speed rises, and increases with higher feedrate or greater contact force.

The surface roughness based on the multivariate regression empirical formula is [40,41].

$$\mathrm{Ra} = \lambda F^{a} v^{b} n^{c} \tag{12}$$

where $\lambda$ is surface roughness coefficient. The values of $a$, $b$, and $c$ are calibrated through above single factor experiments in the subsection 2.2.1. In order to establish a complete surface roughness model, the value of $\lambda$ is calibrated through linear fitting (Fig. 8). Substituting $a$, $b$, $c$, and $\lambda$ into Eq. (12), the surface roughness model is as follows

$$\mathrm{Ra} = 15280 F^{0.410} v^{0.502} n^{-1.178} \tag{13}$$

### 3.2. Surface roughness modeling considering tool wear

The empirical formula for surface roughness shown in Eq. (13) does

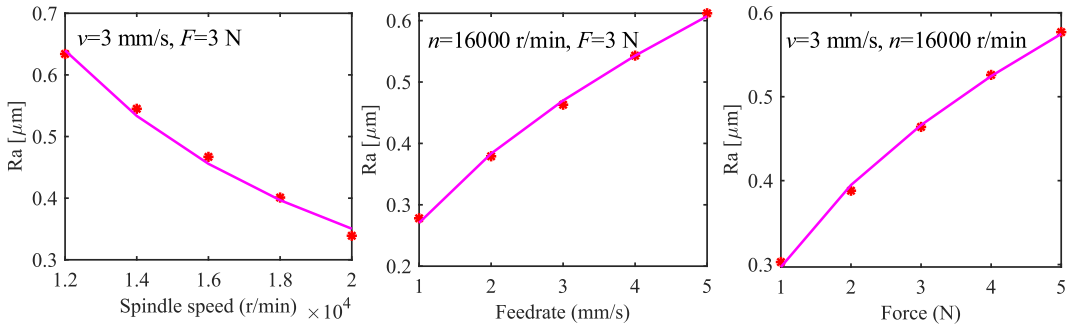



**Fig. 7.** Influence of grinding parameters on surface roughness.

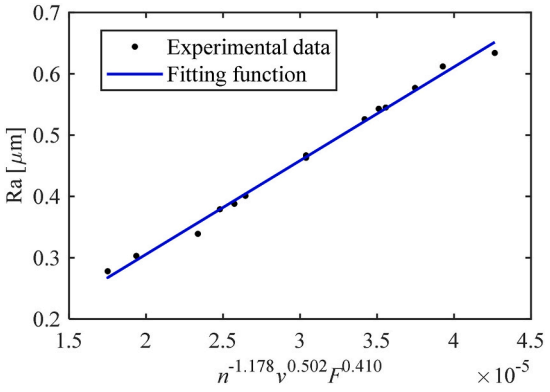



**Fig. 8.** Surface roughness coefficient fitting.

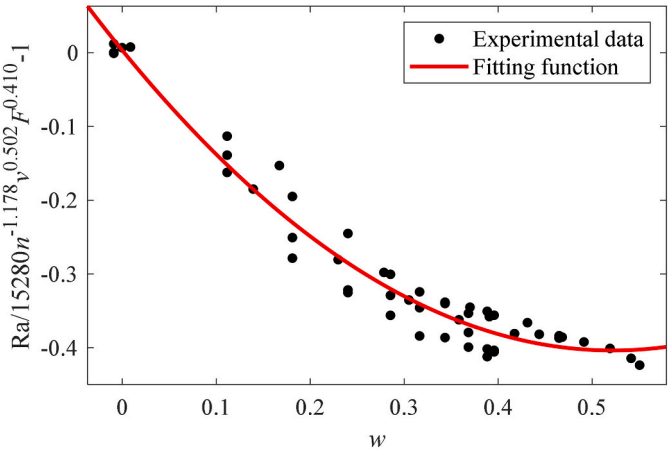



**Fig. 9.** Nonlinear correction function fitting based on tool wear factor.

not incorporate the effect of tool wear. However, the tool with different wear degrees also causes changes in the workpiece surface roughness [25]. Consequently, further investigation into the correlation between surface roughness and tool wear degree is warranted.

To characterize the influence of tool wear on surface roughness, the tool wear degree is substituted into the surface roughness model (Eq. (12)), and a nonlinear correction based on the tool wear degree is introduced as follows

$$\mathrm{Ra} = \lambda F^{a} v^{b} n^{c} (1 + f(w)) \tag{14}$$

where $f(w)$ represents a nonlinear tool wear-dependent correction factor, derived through polynomial regression of experimental grinding data. In order to obtain the accurate form of $f(w)$, grinding experiments are conducted under different contact forces and feedrates, and the workpiece surface roughness at different times is recorded. Fig. 9 displays the function fitted to the experimental data. Based on this, the form of the $f(w)$ is fitted as

$$f(w) = 0.002 - 1.519w + 1.288w^2 + 0.26w^3 \tag{15}$$

To validate the accuracy of the surface roughness model based on tool wear nonlinear correction, multiple grinding tests are conducted, and predicted values based on tool wear nonlinear correction are compared with actual measured values. According to the experimental results (Fig. 10), the model's predicted values of surface roughness agree well with actual experimental measurements, with an $R^2$ value of 0.946, an RMSE value of 0.012 μm, and a MAE value of 0.01 μm, indicating that it can effectively predict surface roughness when considering tool wear. Furthermore, the maximum absolute surface roughness error is about 0.02 μm. Therefore, the correctness of the proposed surface roughness model based on tool wear nonlinear correction is verified.

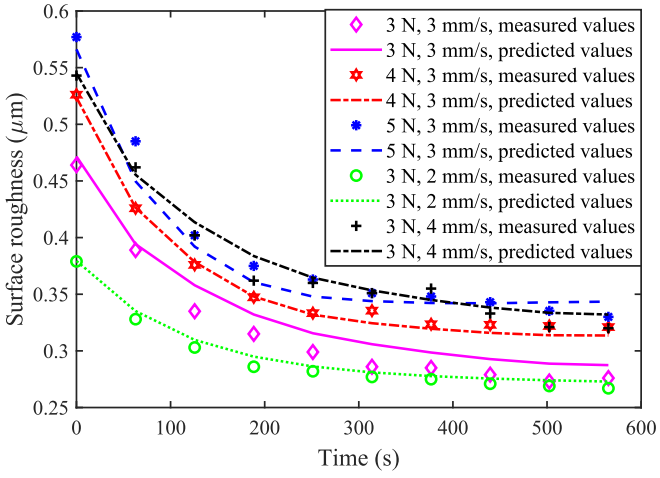



**Fig. 10.** Verification of the surface roughness model based on tool wear nonlinear correction.

## 4. Grinding process optimization with hybrid constraints

In order to ensure required MRD and surface quality of curved parts, the decrease in MRD caused by tool wear must be compensated while meeting the surface quality. This necessitates the optimization of grinding process parameters. This section optimizes the contact force and feedrate with the consideration of tool wear to ensure maximum grinding efficiency while meeting MRD and surface quality.

### 4.1. Constraint analysis in optimization of robot compliant grinding process

For robotic compliant grinding of curved components, this paper employs a 6-DOF robotic system integrated with an end-effector (Fig. 11). In robotic grinding, tool orientation is commonly expressed in *z*-*y*-*z* Euler angles, denoted as $\boldsymbol{\Theta} = [\alpha, \beta, \gamma]^{\mathrm{T}}$ [42]. The tool position $\boldsymbol{P}(\omega)$ and orientation $\boldsymbol{\Theta}(\eta)$ are modeled using B-splines, with $\omega$ and $\eta \in [0,1]$ serving as the spline parameters. Additionally, the grinding paths are further parameterized by arc-length $s$ as [43].

$$[\boldsymbol{P}(\omega(s)), \boldsymbol{\Theta}(\eta(s))]^{\mathrm{T}} = [P_x(\omega(s)), P_y(\omega(s)), P_z(\omega(s)), \alpha(\eta(s)), \beta(\eta(s)), \gamma(\eta(s))]^{\mathrm{T}} \tag{16}$$

The robot joint displacements $\boldsymbol{q}(s)$ are calculated using the inverse kinematics model as follows

$$\boldsymbol{q}(s) = [q_1(s), q_2(s), q_3(s), q_4(s), q_5(s), q_6(s)]^{\mathrm{T}} = IK(\boldsymbol{P}(\omega(s)), \boldsymbol{\Theta}(\eta(s))) \tag{17}$$

where $IK(\cdot)$ is the inverse kinematics function that can be obtained by Denavit-Hartenberg theory [44].

To ensure the required MRD, according to the material removal model considering tool wear established by Eq. (5), the contact force should satisfy

$$F = \left(\frac{h\dot{s}}{k_p(t)n}\right)^2 \left(\frac{r_t r_w}{r_t + r_w}\right) \tag{18}$$

where $\dot{s} = ds/dt$ represents grinding feedrate. During the grinding process, if $F$ is too large, it causes a large deformation of grinding tool, which degrades the grinding quality. Therefore, $F$ should satisfy

$$F \le F_{\max} \Rightarrow \dot{s} \le \frac{k_p(t)n}{h} \sqrt{F_{\max}\left(\frac{1}{r_t} + \frac{1}{r_w}\right)} \tag{19}$$

where $F_{\max}$ represents the maximum limitation of the force.

For ensuring the surface quality after grinding, the surface roughness model considering tool wear established in Eq. (14) should satisfy

$$\mathrm{Ra} = \lambda F^a \dot{s}^b n^c (1 + f(w)) \le \mathrm{Ra}_{\max} \tag{20}$$

where $\mathrm{Ra}_{\max}$ represents the maximum limitation of the surface roughness.

During feedrate optimization, the robot's motion is constrained by both Cartesian and joint space limitations. In Cartesian space, these constraints include the maximum feedrate and acceleration of the tool center point (TCP) during grinding, as defined by

$$\begin{cases} |\dot{s}| = \left|\dfrac{ds}{dt}\right| \le v_{\max} \\ |\ddot{s}| = \left|\dfrac{d\dot{s}}{dt}\right| \le a_{\max} \end{cases} \tag{21}$$

where $v_{\max}$ and $a_{\max}$ represent the feedrate and acceleration maximum limitations, respectively.

The joint-space constraints comprise velocity and acceleration limitations for each robot joint, determined by

$$\begin{cases} |\dot{\boldsymbol{q}}| = \left|\dfrac{d\boldsymbol{q}}{dt}\right| = \left|\dfrac{d\boldsymbol{q}}{ds}\dfrac{ds}{dt}\right| \le \dot{\boldsymbol{q}}_{\max} \\ |\ddot{\boldsymbol{q}}| = \left|\dfrac{d^2\boldsymbol{q}}{ds^2}\left(\dfrac{ds}{dt}\right)^2 + \dfrac{d\boldsymbol{q}}{ds}\dfrac{d^2s}{dt^2}\right| \le \ddot{\boldsymbol{q}}_{\max} \end{cases} \tag{22}$$

where $\dot{\boldsymbol{q}}_{\max}$ and $\ddot{\boldsymbol{q}}_{\max}$ are the velocity and acceleration maximum limitations of the robot joints.

### 4.2. Process optimization under hybrid constraints

The objective of grinding process optimization is to ensure maximum processing efficiency while meeting MRD and surface quality considering tool wear, and grinding feedrate constraints considering robot dynamics. The time optimal feedrate planning model for the robotic compliant grinding is expressed as

$$\text{minimize } t = \int_0^{S_\Sigma} \frac{ds}{\dot{s}} \tag{23}$$

where $t$ represents the total grinding time, $s$ represents the arc-length of grinding path, and $S_\Sigma$ represents the total arc-length.

To solve the optimization problem in Eq. (23), we establish a mapping between arc length $s$ and feedrate $\dot{s}$. Leveraging B-spline's superior local adjustment capability, the feedrate profile is respresented using a cubic ($p = 3$) B-spline curve as follows

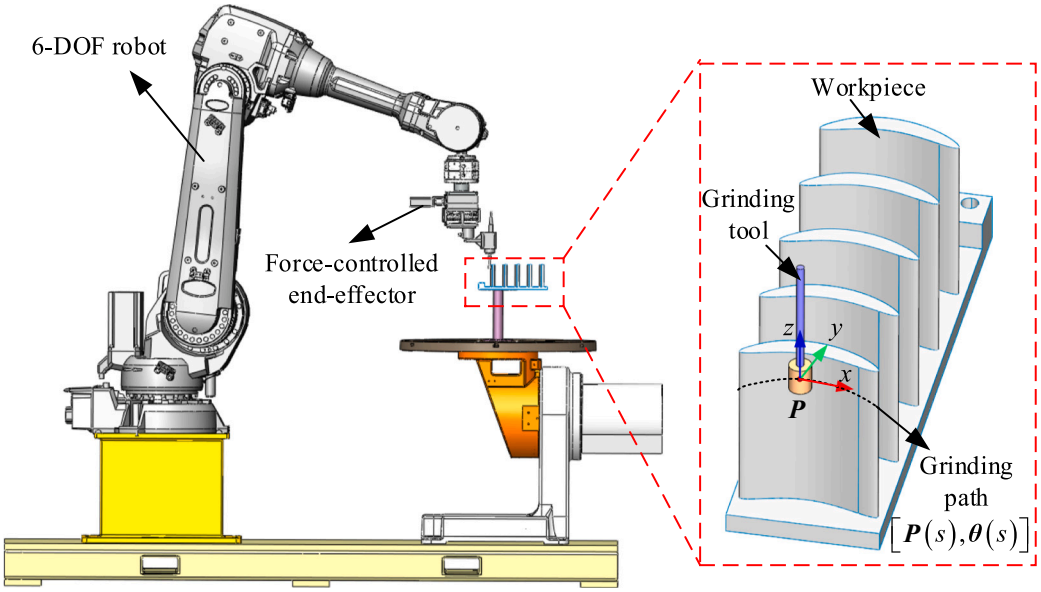



**Fig. 11.** Robotic grinding system for complex curved components.

$$\dot{s}(u) = \sum_{0}^{n-1} B_{i,p}(u) f_i \tag{24}$$

where $B_{i,p}(u)$ represents the B-spline basis function, $f_i$ denotes the control points, and $n$ indicates the number of control points determined by equally dividing the path into $n$-1 segments with arc length $\Delta s$. The parameter $u \in [0,1]$ is normalized by $u = s/S_\Sigma$.

During the process parameter optimization, the tool wear degree $w(t)$ is updated by the following formula

$$w(t) = w(t-1) + r(t) \cdot \frac{\Delta s}{\dot{s}} \tag{25}$$

Then the real-time material removal depth coefficient $k_p(t)$ is

$$k_p(t) = (1 - w(t))k_{p0} \tag{26}$$

Based on Eqs. (19)–(26), the optimization function for maximizing grinding efficiency while satisfying MRD and surface quality considering tool wear, and grinding feedrate constraints considering robot dynamics of the force-controlled compliant grinding system is formulated as

$$\text{minimize } t = \int_0^{S_\Sigma} \frac{ds}{\dot{s}}$$

$$\text{s.t.} \begin{cases} |\dot{s}| \le v_{\max}, |\ddot{s}| \le a_{\max} \\ |\dot{\boldsymbol{q}}| \le \dot{\boldsymbol{q}}_{\max}, |\ddot{\boldsymbol{q}}| \le \ddot{\boldsymbol{q}}_{\max} \\ \dot{s} \le \dfrac{k_p(t) n}{h} \sqrt{F_{\max}\left(\dfrac{1}{r_t} + \dfrac{1}{r_w}\right)} \\ \lambda F^a \dot{s}^b n^c (1 + f(w)) \le \mathrm{Ra}_{\max} \end{cases} \tag{27}$$

Eq. (27) presents a nonlinear optimization problem that is efficiently resolved through the SQP method utilizing the fmincon solver in MATLAB [45].

Upon optimizing the feed profile $\dot{s}$, the tool tip displacement $s$ is then derived through feedrate integration, enabling calculation of the force command $F$ at different positions on the workpiece based on the MRD constraint in Eq. (18).

## 5. Experiments

### 5.1. Experiment setup

The experimental setup (Fig. 12) consists of a compliant force-controlled end-effector mounted on the ABB 4600–60 robot flange and a workpiece secured to the worktable. The system operates through coordinated control: the robot executes trajectory tracking, while the compliant end-effector maintains prescribed force (the control cycle is 4 ms). The base of tool is rubber and the abrasive is diamond, and the material of the workpiece is TC4.

Fig. 13 displays the workpiece model and planned grinding path used in the grinding process optimization with hybrid constraints. The desired $h$ is set to 0.01 mm, the spindle speed $n$ is set to 16,000 rpm, and the grinding tool radius is 4 mm. The Cartesian space dynamic constraints of the robot are set to $v_{\max} = 4$ mm/s and $a_{\max} = 10$ mm/s$^2$, while joint constraints are $\dot{\boldsymbol{q}}_{\max} = [100, 100, 100, 40, 40, 40]^\circ/s$ and $\ddot{\boldsymbol{q}}_{\max} = [300, 300, 300, 150, 150, 150]^\circ/s^2$.

### 5.2. Experimental validation

To validate the efficacy and advantages of the proposed method in ensuring both material removal accuracy and surface roughness, a comparison method is chosen where grinding parameters are planned by the MRD model that does not consider tool wear, and there is no surface roughness constraint. For a fair comparison, the robot joint and Cartesian space limitations, desired material removal depth, and spindle speed are the same in different situations.

Fig. 14 shows the robot feedrate and contact force planned by the

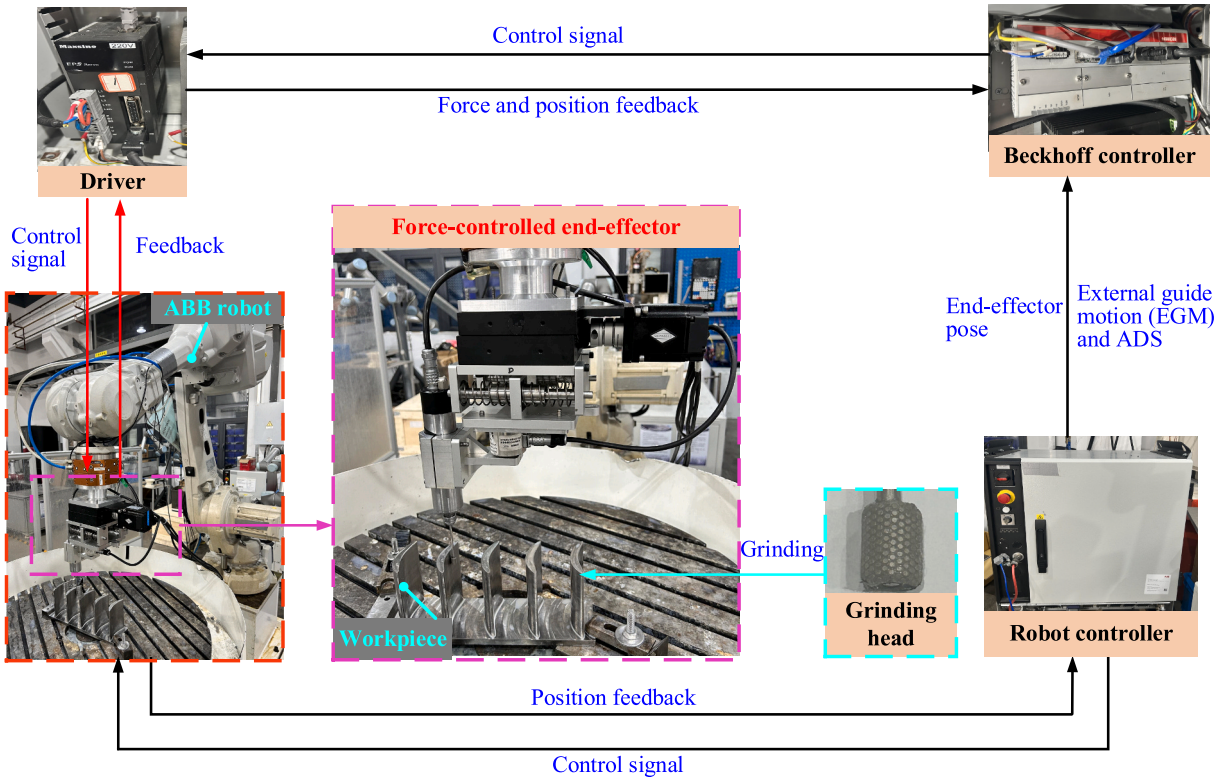



**Fig. 12.** Robotic compliant grinding experiments on the curved part.

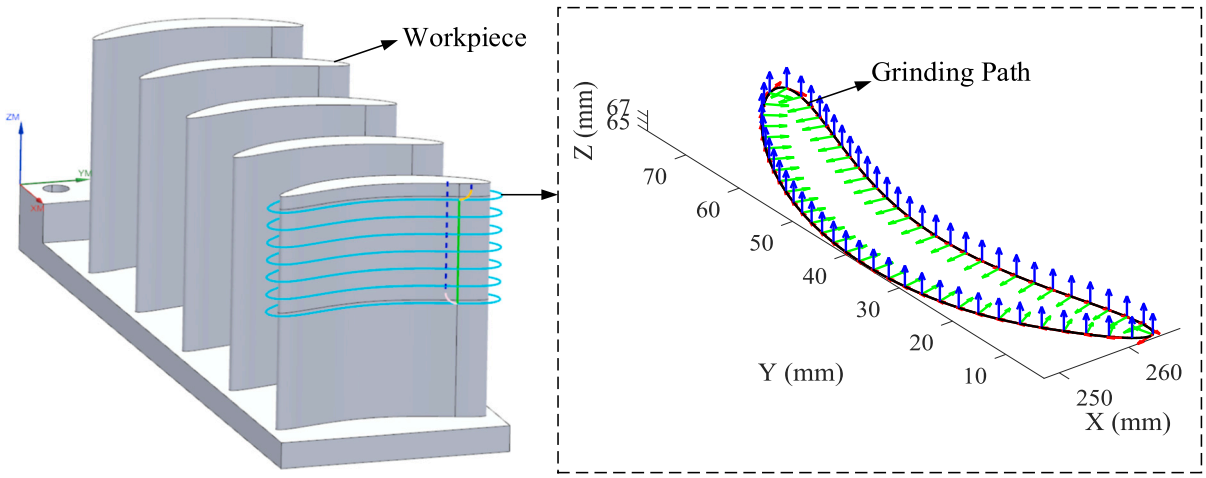



**Fig. 13.** Workpiece model and the grinding path.

proposed method and the comparison method. Then, based on the planned feedrate and force, the MRD and surface roughness are calculated using Eq. (5) and Eq. (14), respectively, as shown in Fig. 15.

The calculation results in Fig. 15 confirm the superior capability of the proposed method in simultaneously ensuring MRD and surface roughness, while the comparison method does not satisfy the MRD and surface roughness constraints. The grinding time of proposed method and comparison method are 461.8 s and 455.7 s, respectively. This proves that the proposed method improves material removal accuracy and surface quality compared with the comparison method when grinding efficiencies are roughly the same. Therefore, it shows the effectiveness of the proposed method in process parameter planning.

Fig. 16 shows the normalized ratios of joint velocity and acceleration relative to their operational limits, demonstrating that the proposed method successfully maintains joint velocity and acceleration within their prescribed safety boundaries throughout the robotic motion.

Fig. 17 presents the force tracking performance during grinding experiments, which respectively show the actual forces tracking and their tracking errors. Experimental results indicate that both the proposed and comparison methods can effectively track the planned grinding contact force, and the force tracking errors are all about ±0.4 N. In the robotic grinding experiment of curved parts, the force-controlled end-effector adopted a Beckhoff controller, and the position of the robot end-effector in the workpiece coordinate system is obtained through External Guide Motion (EMG) and Automation Device Specification (ADS) communication with a sampling period of 4 ms. Then, the arc-length of the grinding trajectory is calculated by numerically integrating the incremental displacement of the end-effector.

The robot feedrate tracking performance is illustrated in Fig. 18, depicting the actual feedrate tracking and corresponding errors, respectively. The results demonstrate that both the proposed and comparison methods successfully track the desired grinding feedrate, with feedrate tracking errors consistently around ±0.45 mm/s.

For further explorations, the contours of complex curved parts before and after grinding are measured by a coordinate measuring machine (Fig. 19), and the MRD errors are calculated. In addition, the ground workpieces roughness is measured using a surface roughness tester.

The efficacy of the proposed process parameter optimization method considering tool wear is verified by measuring contours 1 to 4. Contour 1 and contour 2 correspond to the first and seventh grounds of grinding of the part using the proposed method, respectively. Contour 3 and contour 4 correspond to the first and seventh grounds of grinding of the part using the comparison method, respectively. Fig. 20 shows that the maximum MRD errors of contours 1, 2, 3, and 4 are 3.1 μm, 3.2 μm, 3.7

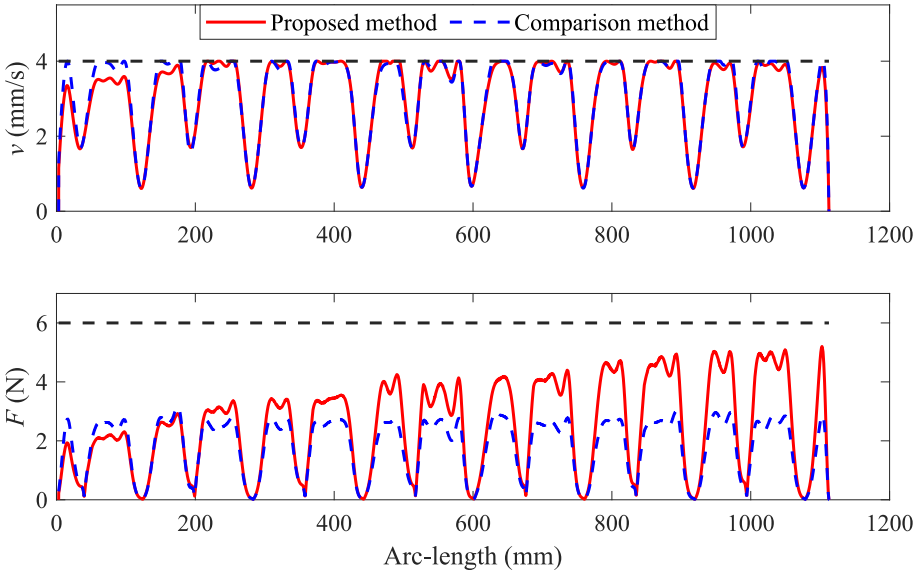



**Fig. 14.** Planned results of grinding feedrate and force.

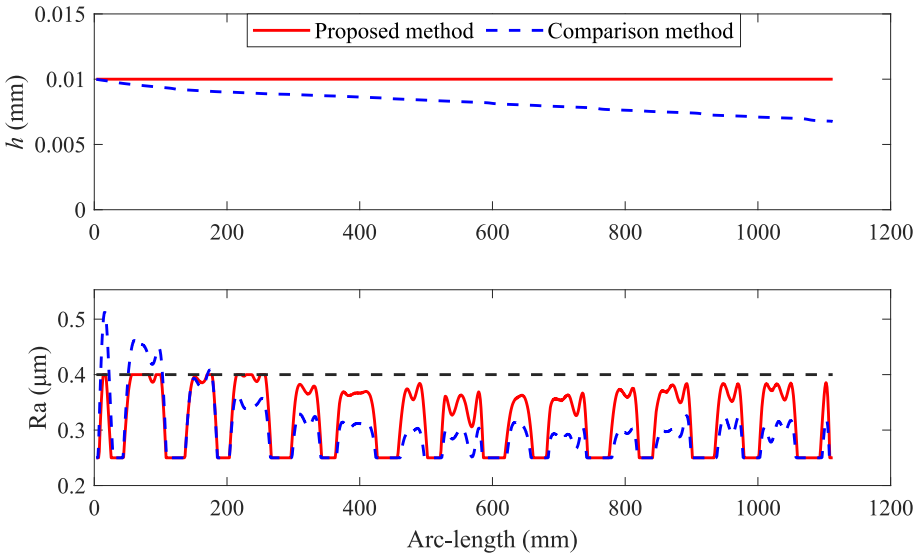



**Fig. 15.** Calculation results of the MRD and surface roughness.

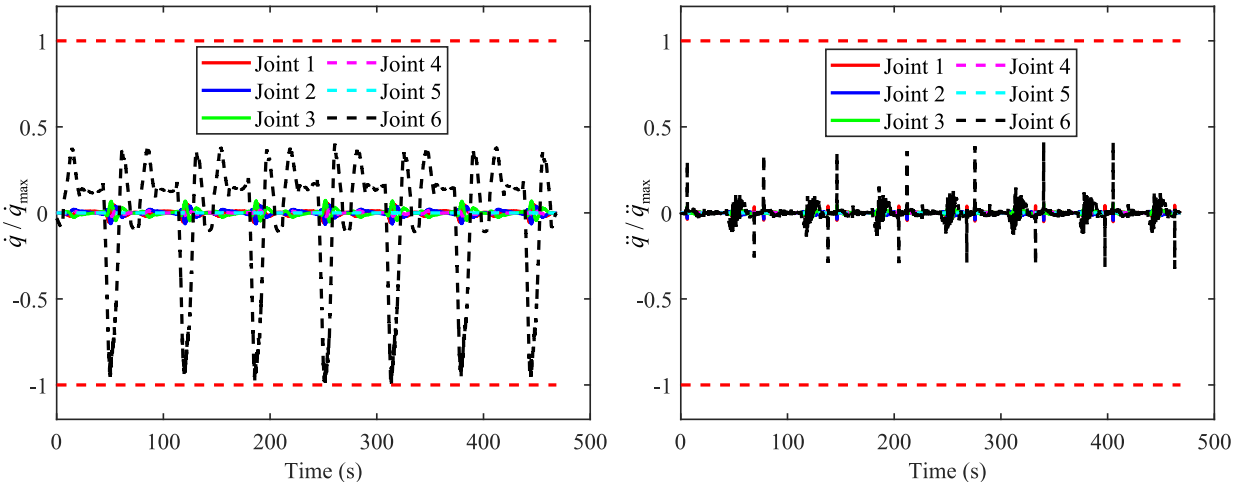



**Fig. 16.** Normalized ratios of robot joint velocity and joint acceleration relative to constraints.

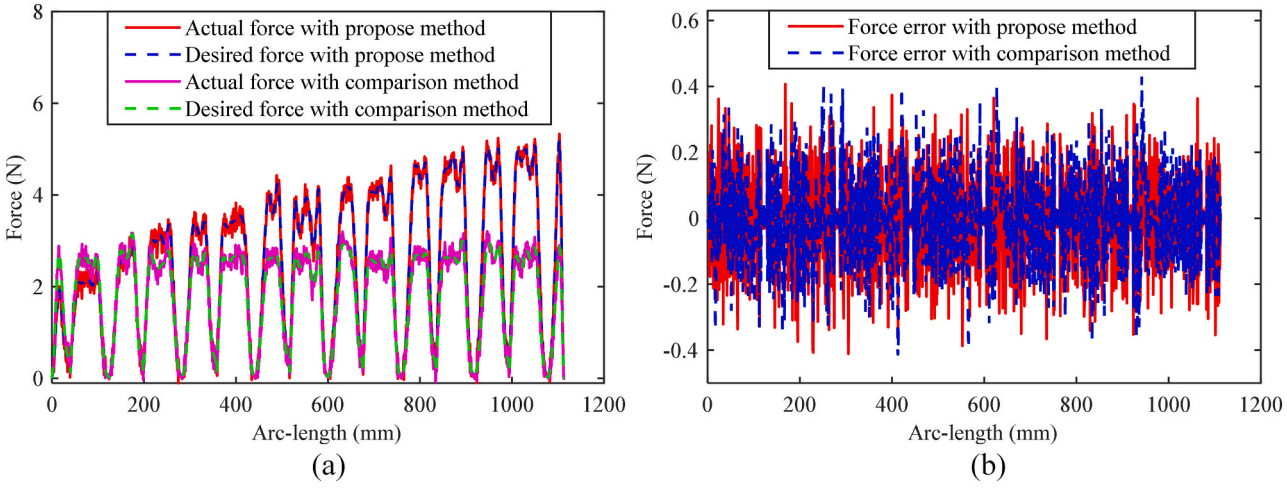



**Fig. 17.** Comparison results of the grinding forces: (a) planned and actual forces; (b) force tracking errors.

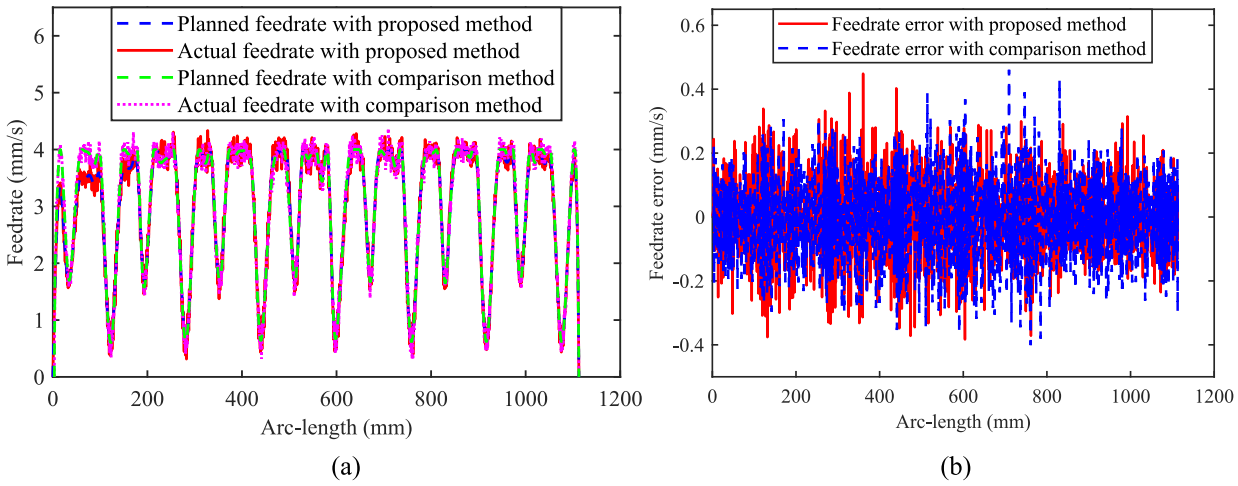



**Fig. 18.** Comparison results of the feedrates: (a) planned and actual feedrates; (b) feedrate tracking errors.

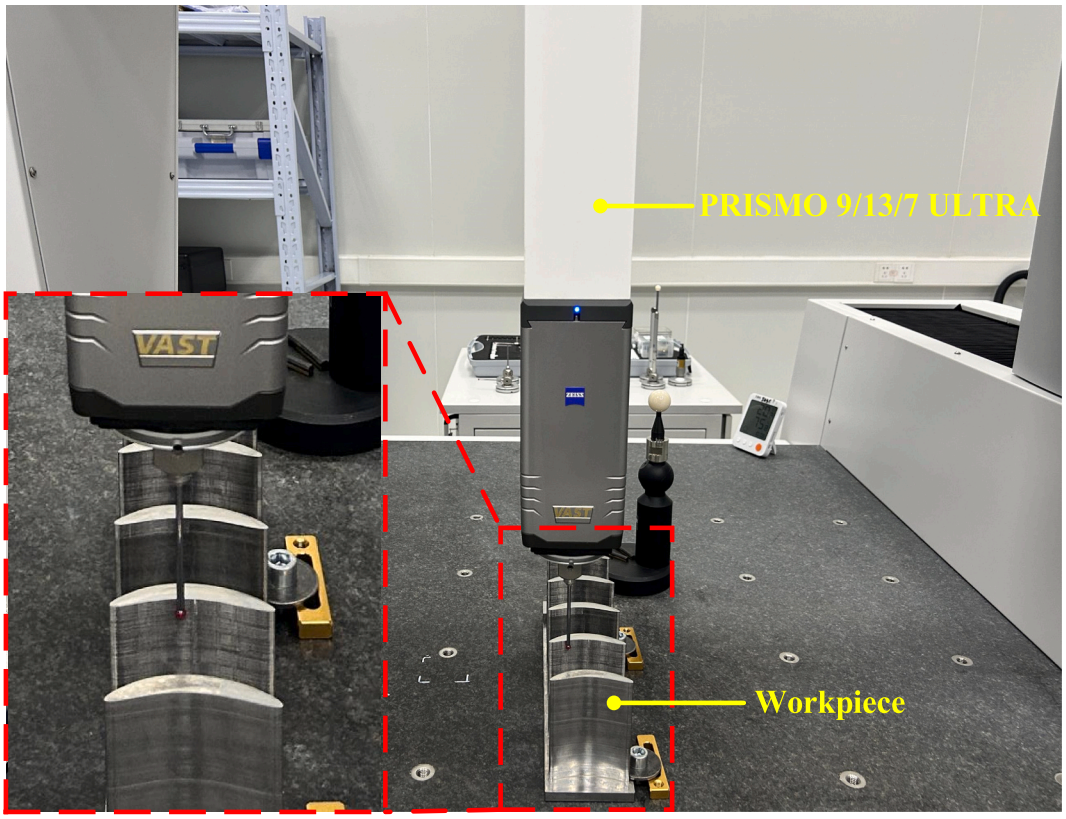



**Fig. 19.** Workpiece Contour Measurement Schematic.

μm, and 6.7 μm, respectively; the average absolute MRD errors are 0.9 μm, 1.0 μm, 1.2 μm, and 3.6 μm, respectively.

Measurement results of contours 1 and 2 indicate that the maximum and average absolute MRD errors are basically the same for the first and seventh grounds after the workpiece are ground using the proposed method. The proposed process optimization method can compensate for the reduction in MRD caused by tool wear by increasing the grinding contact force, thereby ensuring the consistency of MRD. Measurement results of contours 1 and 3 show that in the initial stage of the grinding process, the influence of tool wear on MRD is relatively limited, causing only slight changes. Measurement results of contours 2 and 4 reveal that with the increase of grinding time, in the seventh ground of grinding of the parts using the comparison method, the MRD caused by the wear of the grinding tool decreases significantly, and the average absolute MRD error is 3.6 μm; while the proposed method can compensate for the reduction in MRD caused by tool wear by increasing the grinding contact force, ensuring the consistency of MRD.

In summary, the process parameters planned by the proposed method effectively compensate for the reduction in MRD caused by tool wear, which ensure the consistency of MRD and improve contour accuracy.

Fig. 21 reveals the comparison of surface roughness after grinding using the proposed method and the comparison method. The average surface roughness of the first and seventh grounds using the proposed method is Ra = 0.390 μm and Ra = 0.361 μm, respectively. The average surface roughness of the first and seventh grounds using the comparison

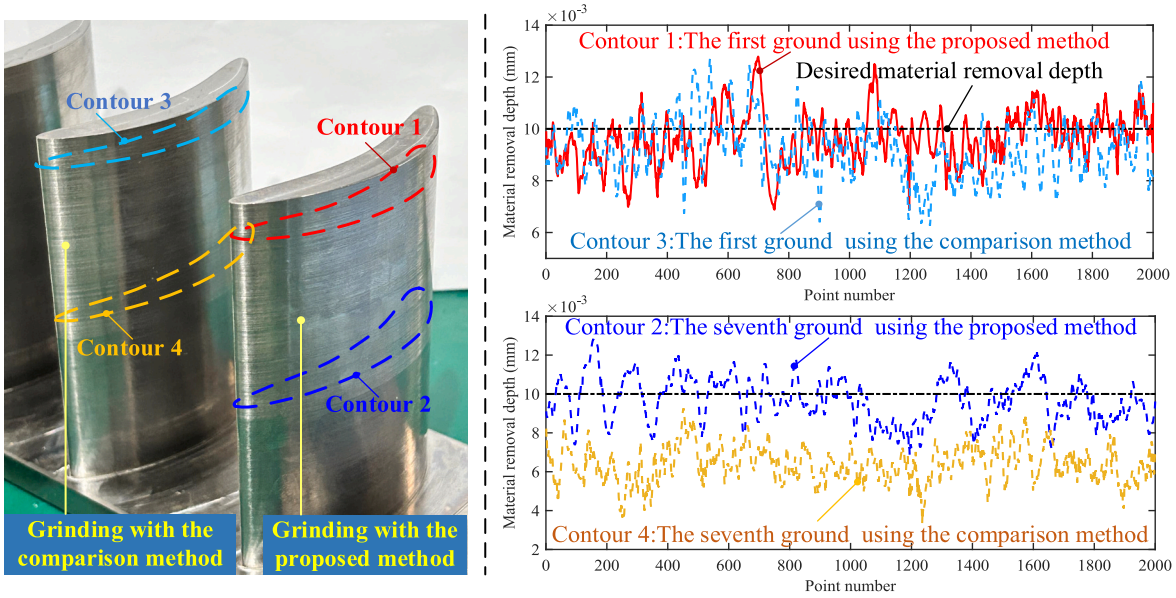



**Fig. 20.** Comparison of MRD after grinding.

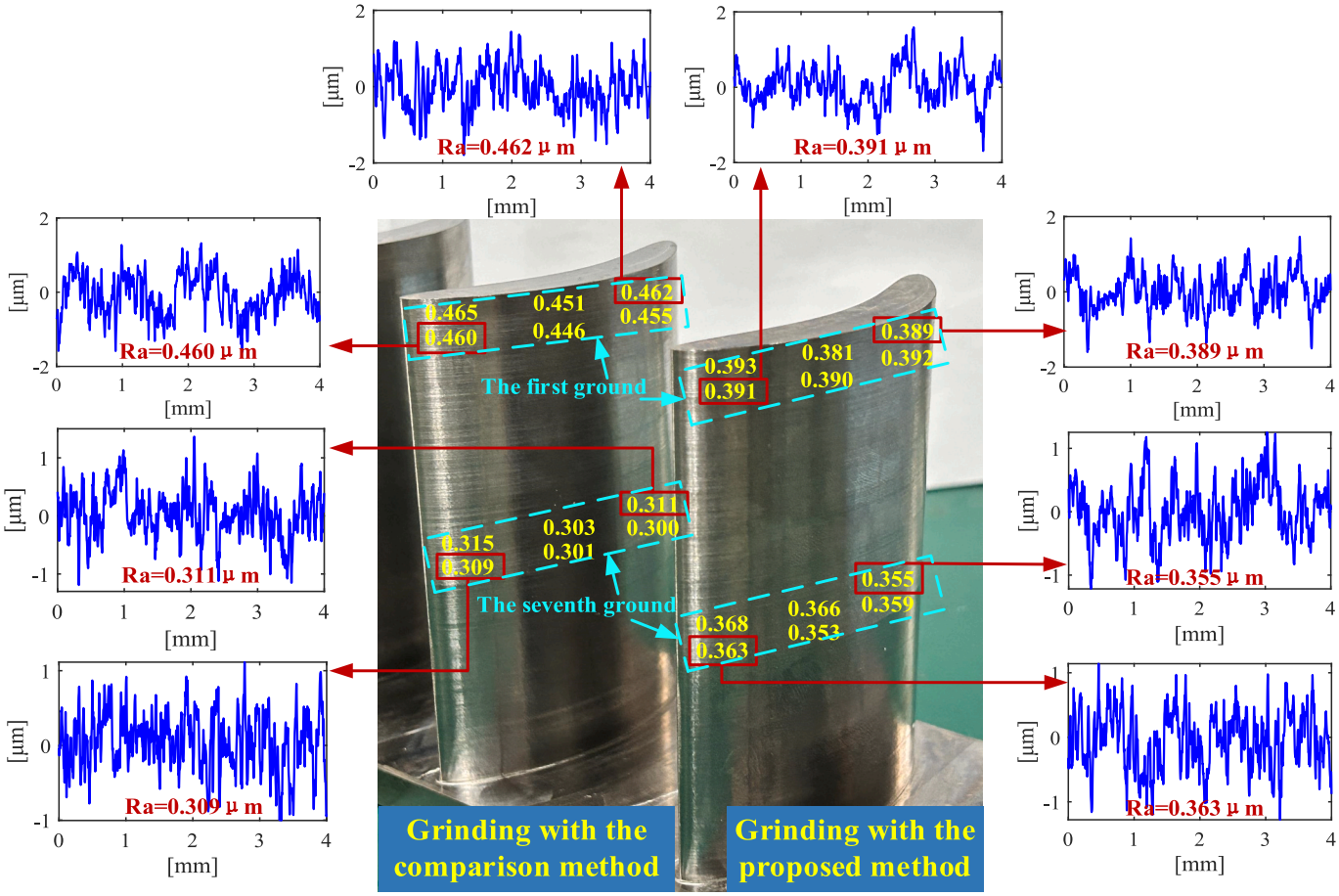



**Fig. 21.** Comparison of surface roughness after grinding.

method is Ra = 0.456 μm and Ra = 0.307 μm, respectively. Therefore, the surface roughness after grinding using the comparison method cannot always be guaranteed to be within the limited surface roughness range due to the lack of surface roughness constraint, while the surface roughness can always be guaranteed to be within the limited surface roughness range by constraining the surface roughness limitation when optimizing grinding parameters with the consideration of tool wear using the proposed method.

It is precisely because the proposed process parameter optimization method can effectively compensate for the MRD error caused by tool

wear under the condition of satisfying the surface roughness constraint, the proposed method has better grinding performance regarding contour accuracy and surface quality of curved parts.

## 6. Conclusion

In this paper, a robotic grinding parameter optimization approach considering tool wear to guarantee material removal accuracy and surface quality is proposed. The MRD coefficient under different process parameters at different times is predicted by the multivariate Gaussian process regression method, and a material removal model considering tool wear is established to achieve accurate quantification of tool wear. To characterize the influence of tool wear on surface roughness, a surface roughness prediction model is constructed in combination with the tool wear degree. A process parameter optimization method is proposed to compensate for the MRD error caused by tool wear while meeting the surface roughness constraint. Experimental results show that compared with the robot grinding process parameter optimization method without considering tool wear and surface roughness constraint, the proposed method can effectively compensate for the MRD error caused by tool wear while meeting the surface roughness constraint, and the maximum MRD error and maximum roughness are reduced by 53.7 % and 15.5 %, respectively. The proposed process parameter optimization method improves the material removal accuracy and surface roughness of curved surface parts after grinding. Future research could focus on material removal modeling and process parameter optimization for irregularly shaped grinding heads to achieve high-precision grinding of more complex curved parts.

## CRediT authorship contribution statement

**Haiqing Chen:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Jixiang Yang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Qi Qi:** Validation, Methodology, Formal analysis, Data curation. **Han Ding:** Writing – review & editing, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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